

Motivation

LLMs adopt personas when prompted, and these behavioral shifts correspond to recoverable directions in activation space [1, 2]. Contrastive steering can manipulate outputs along these directions [3, 4, 7], though instruction-level persona control remains unstable across dialog turns [8]. Persona features have been linked to emergent misalignment [5].

The Persona Selection Model [9] proposes that LLMs learn a structured persona space during pre-training. **Can contrastive extraction recover this structure?**

We extract role vectors for **275 roles** from OLMo3-7B-Instruct and analyze the geometry of the resulting persona space, finding interpretable structure beyond the previously documented assistant axis [1].

Method

For each role, we generate paired responses: one with a role-specific system prompt and one with no system prompt. An LLM judge scores both for role alignment (0–100); only pairs passing a quality threshold are retained.

We compute contrastive activation vectors at layer 16:

$$\vec{v}_{\text{role}} = \vec{a}_{\text{prompted}} - \vec{a}_{\text{baseline}}$$

These 4,096-dim vectors encode the persona direction in activation space.

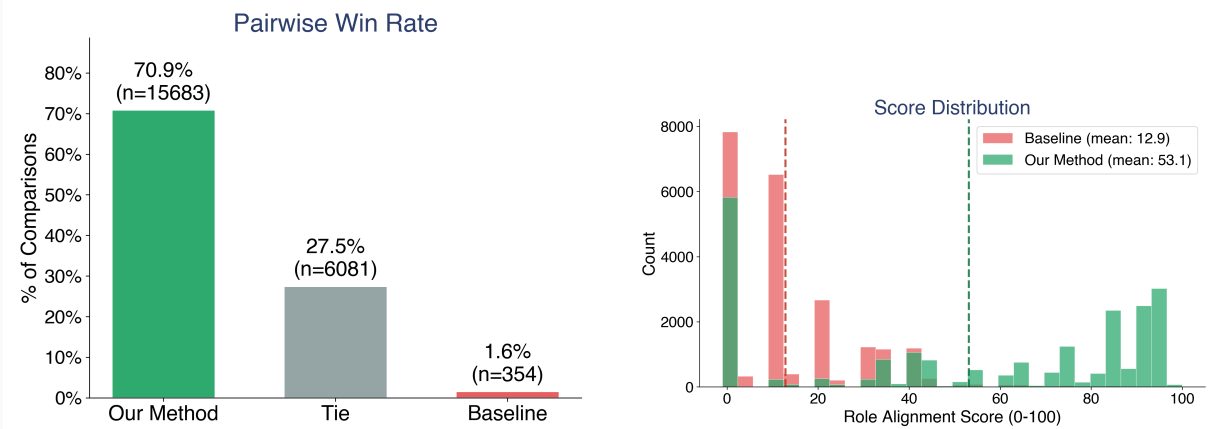
	Lu et al.	Ours
Extraction	Mean act.	Contrastive diff
Filter	None	LLM judge
Questions	230+	20–50
Behavior	Assistant-like	1st-person role
Eval	Not done	5-axis judge
Mean sub.	Yes	No

Model: OLMo3-7B-Instruct | **Layer:** 16 | **Judge:** GPT-4.1

Roles: 275 extracted, 97 evaluated | **Scale:** 500k+ Q/A pairs

Empirical Results

Steered outputs and the Assistant Axis baseline [1] are each compared against a prompt-engineered gold persona response using a 5-axis GPT-4.1 judge: *emotional register*, *vocabulary*, *social dynamic*, *motivation*, *worldview*. A/B ordering randomized; results debiased.



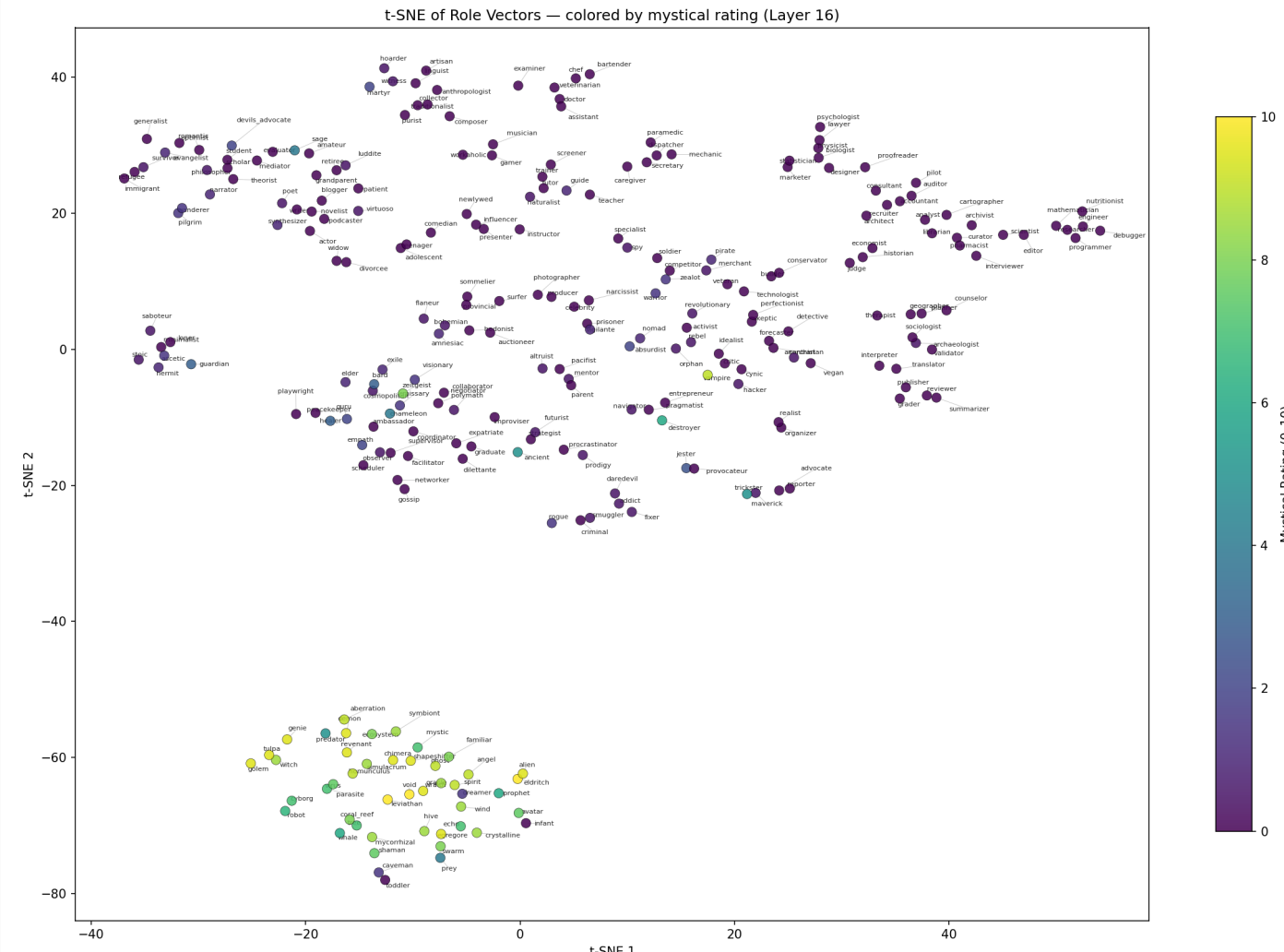
70.9% win rate across 22,118 comparisons (97 roles). Mean role-alignment score: **53.1** (ours) vs. **12.9** (baseline). We lead on all 97 evaluated roles with zero losses.

The Fantastical Separation

t-SNE visualization of our 275 role vectors reveals two geometrically distinct clusters that were not anticipated by our extraction methodology.

The upper cluster contains **human/realistic roles**: *doctor, lawyer, pirate, mechanic, historian, economist, comedian, spy, facilitator, judge*. The lower cluster contains **fantastical and non-human roles**: *avatar, eldritch, alien, angel, mystic, symbiote, shaman, whale, coral reef, revenant*. Notably, *toddler* and *infant* cluster with this group, suggesting the model treats pre-linguistic entities as distinct from adult conversational personas.

To validate this separation, we independently rated each role’s “mystical-ness” (0–10) using a separate LLM evaluator. The resulting scores align with the cluster boundaries, suggesting the separation reflects meaningful semantic structure.

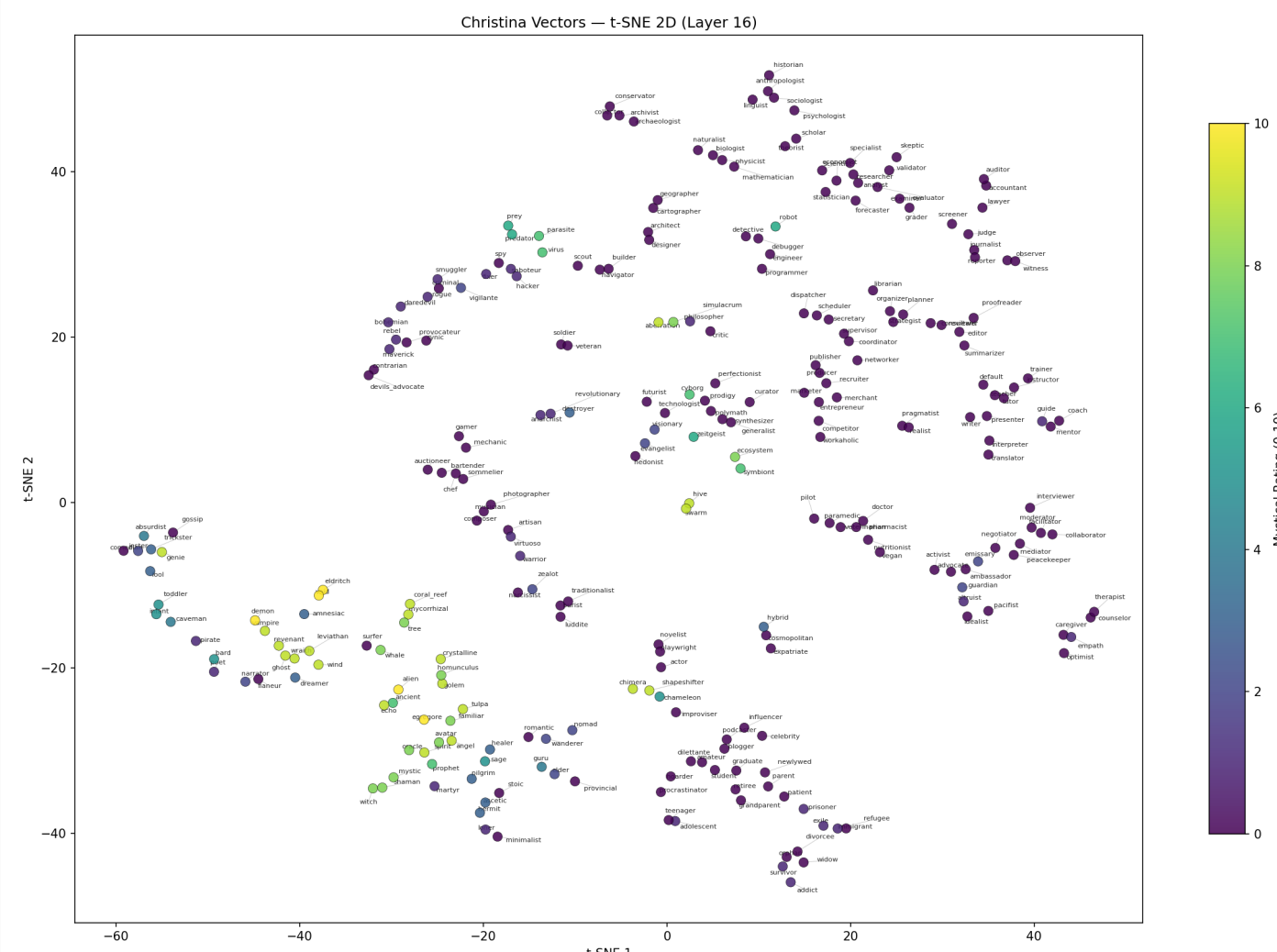


t-SNE of 275 role vectors (layer 16), colored by independently-rated mystical score (0–10). The lower cluster contains fantastical and non-human roles. Perplexity = 5.

Baseline: No Fantastical/Realistic Separation

We replicate the Assistant Axis methodology [1] and apply the same t-SNE visualization. The baseline vectors—extracted via mean activation with mean subtraction—do **not** exhibit analogous cluster separation. Roles are distributed without a clear fantastical/realistic divide.

Both methods operate in the same 4,096-dim activation space. Our contrastive extraction with judge filtering recovers role-specific structure that mean-activation methods do not.

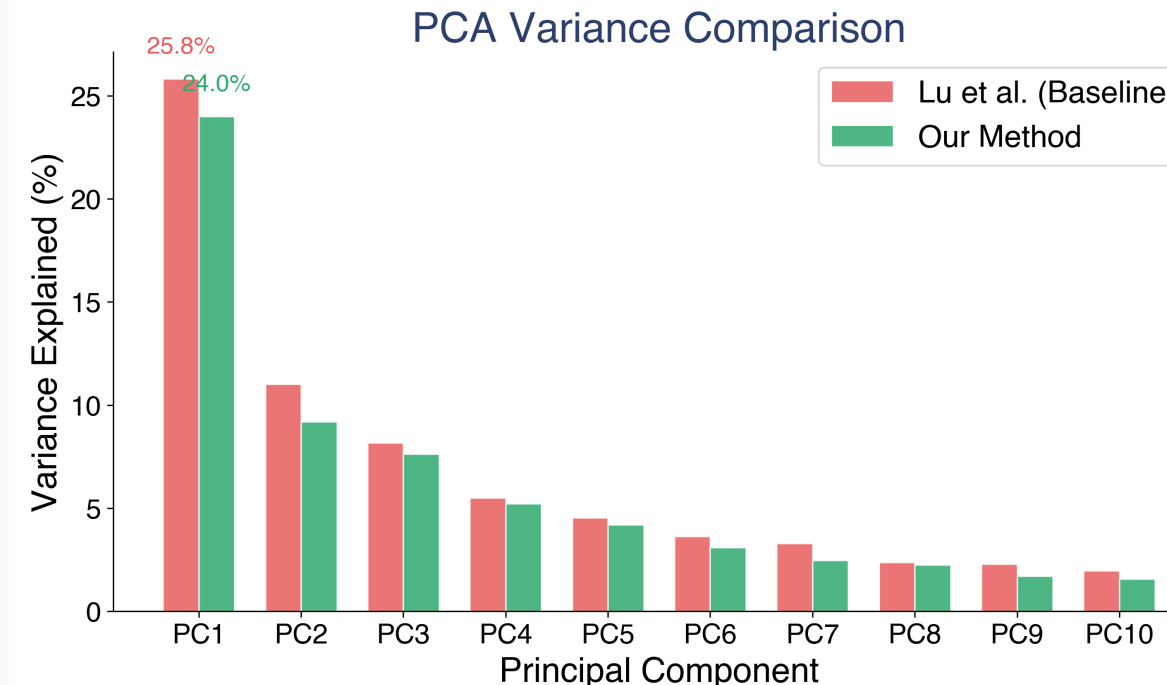


Baseline t-SNE (Lu et al. [1]), colored by the same mystical rating scale as our t-SNE above. Mystical roles (yellow/green) are scattered throughout with no cluster separation. Same roles, same t-SNE parameters.

Geometric Analysis

PCA on the role vector matrix reveals low-dimensional structure: ~ 20 principal components capture 90% of variance across 275 roles in 4,096-dim space.

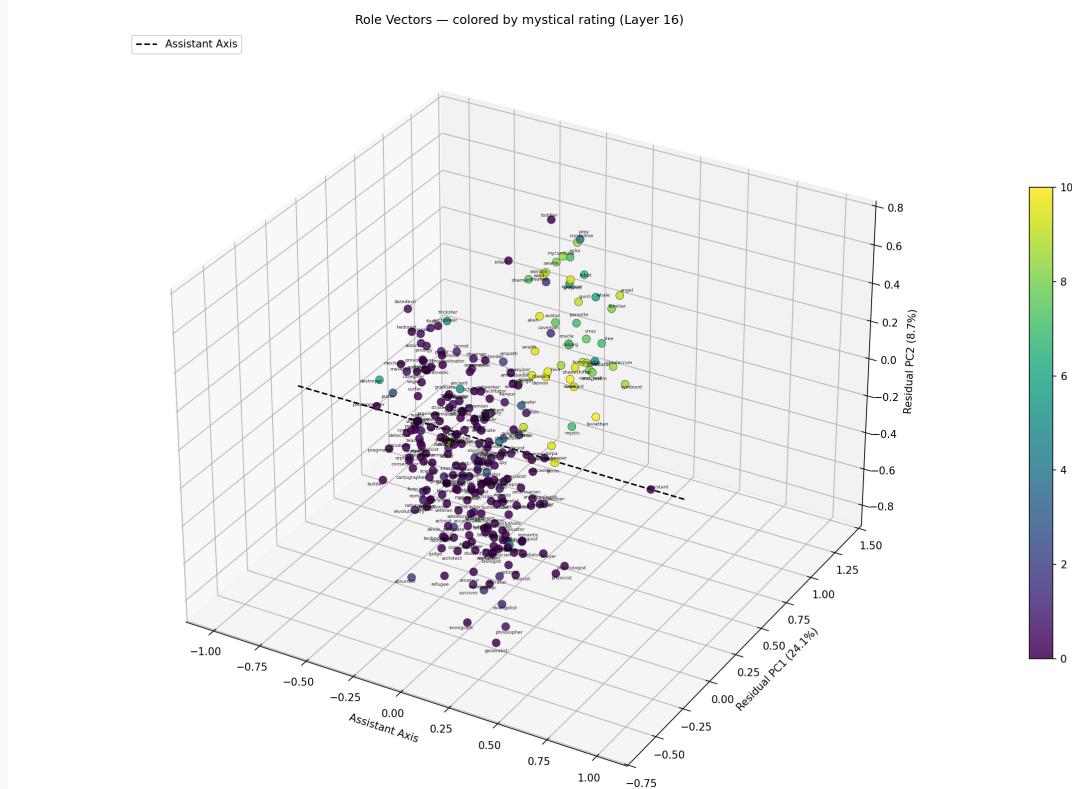
PC1 (associated with the assistant axis [1]) explains 24.0% of variance in our vectors vs. 25.8% in the baseline. Our method distributes variance more evenly across components, consistent with greater inter-role separability.



Variance explained per principal component. Lower PC1 concentration indicates greater separability between roles. ~ 20 PCs needed for 90% cumulative variance (first 10 shown).

PCA Supports Cluster Structure

To test whether the t-SNE finding persists in the raw geometry, we construct a custom 3D basis: the assistant axis as dimension 1, with two residual PCA components as dimensions 2 and 3. High-mystical roles occupy a distinct region along the residual axes, indicating the cluster is present beyond t-SNE.



3D PCA colored by mystical rating. The assistant axis (dashed) defines dimension 1. Fantastical roles (yellow/green) cluster in the upper region of the residual space, separated from realistic roles.

Vector Arithmetic

Role vectors support algebraic composition analogous to word embedding arithmetic [6], suggesting the space encodes relational structure between personas:

warrior – **stoic** + **pacifist**
 $\approx$ *activist, evangelist, coordinator*
scientist – **critic** + **criminal**
 $\approx$ *smuggler, hacker, detective*

Nearest neighbors by cosine similarity. These results suggest compositional structure but require further systematic validation.

Conclusions & Future Work

- Contrastive extraction with judge filtering** wins 70.9% of comparisons against the Assistant Axis baseline [1] across 97 roles (22,118 comparisons, 500k+ Q/A pairs), with mean score 53.1 vs. 12.9. Zero role-level losses.
- Fantastical/non-human roles** occupy a geometrically distinct subspace, visible in both t-SNE and PCA, with boundaries aligned to an independent mystical rating. Pre-linguistic roles cluster with this group.
- Role space exhibits **low-dimensional structure** (~ 20 PCs for 90% variance) and suggestive algebraic compositionality. Our fantastical/realistic separation mirrors the PC1 axis found across three other models [1], providing converging evidence for the Persona Selection Model [9].
- Future work:** Full 275-role evaluation, k-means verification, cross-model generalization [5].

References

[1] Lu et al. (2026). *The Assistant Axis*. [2] Chen et al. (2024). *From Persona to Personalization*. [3] Panickssery et al. (2023). *Steering via Contrastive Activation Addition*. [4] Zou et al. (2023). *Representation Engineering*. [5] Wang et al. (2025). *Persona Features Control Emergent Misalignment*. [6] Wang et al. (2023). *RoleLLM*. [7] Tan et al. (2024). *Generalization of Steering Vectors*. [8] Li et al. (2024). *Instruction (In)Stability in LM Dialogs*. [9] Anthropic (2026). *The Persona Selection Model*.

Acknowledgments

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